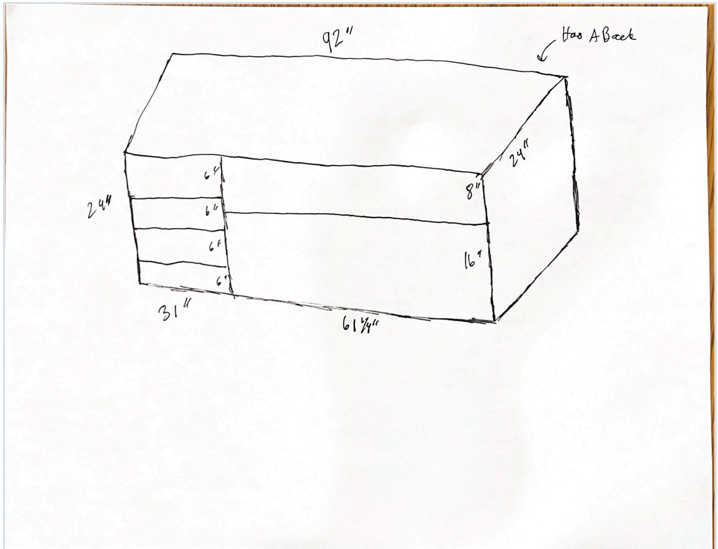
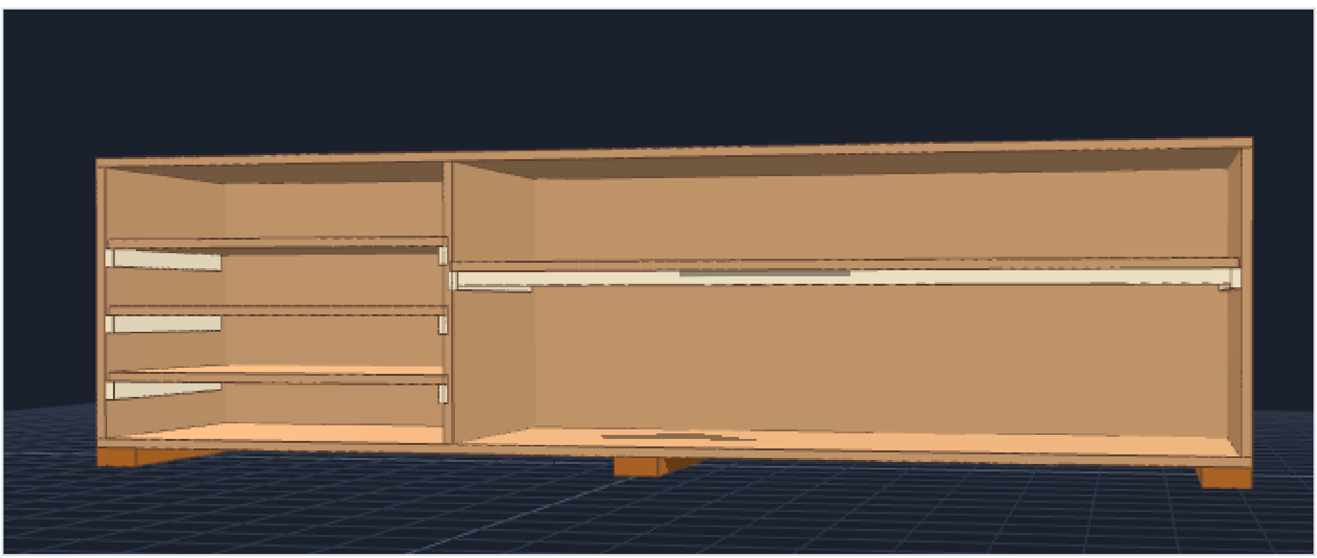
Chemistry Lab Floor Storage Shelf (Shelf A)

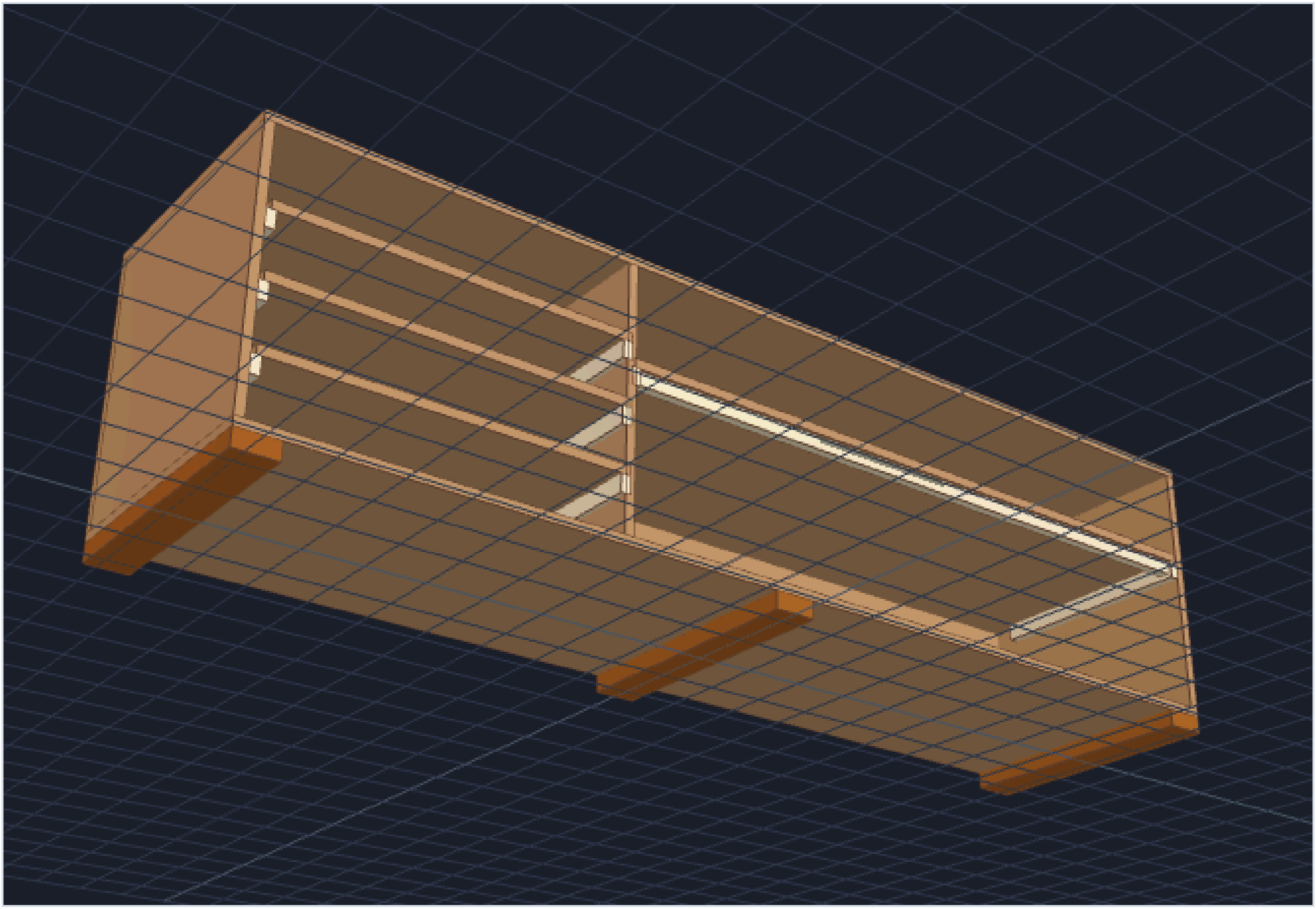


Complete Step-by-Step DIY Build & Material Guide

Front View



View at an angle showing stiffener bars, cleats, and base skids



Interactive 3D model

Open Interactive 3D Floor Shelf Model

> **Interactive 3D View:** Open 360 degrees Floor Shelf Viewer in Browser

Key Specifications

- **Overall Dimensions:** 92″ Wide x 23.25″ Deep x 25.5″ High (24″ Body + 1.5″ Base Skids [P11])
- **Design Features:** Easy cleat assembly [P10], anti-sag long shelf reinforcement [P9], base elevation skids against damp floors [P11], dual-fastened shelves [P7, P8], and chemical-resistant finishing for lab use.
- **Wall Mounting Restriction: NO WALL MOUNTING ALLOWED.** This unit is engineered to be strictly free-standing. Structural stability, squareness, and anti-racking strength are provided by internal side cleats [P10], base skids [P11], and the full 3/4″ rear backing panel [P3].

> **NOTE**
> Do not anchor or screw this shelf into room walls. It must remain fully free-standing. Always build on a flat surface, verify diagonal squareness, and ensure the back panel [P3] is glued and fastened securely to lock the cabinet rigid.

1. Materials to Buy

Lumber & Plywood

Qty	Item	Use
3 Sheets	3/4″ x 4′ x 8′ Sanded Plywood (BCX or Sandeply Birch)	Outer frame (P1, P2, P4, P5, P6), back panel (P3), and shelves (P7, P8)
3 Boards	1″ x 2″ x 8′ Select Pine Board	Long shelf stiffener lip (P9) and side support cleats (P10)
1 Board	2″ x 4″ x 8′ Construction Pine Board	Base skids / feet underneath unit (P11)

> **Pro Scout Tip:** Ask the store associate at the Home Depot or Lowe's lumber panel saw to rip your 4′ x 8′ sheets into the 23.25″ and 24.0″ wide strips for free or cheap. This makes transporting the wood much easier and guarantees straight cuts!

Fasteners, Hardware & Glue

Qty	Item	Use
1 Bottle	Titebond II Wood Glue (16 oz)	Primary structural bonding across all wood-to-wood joints (80%+ total strength)
1 Box	2″ #8 Star-Drive Wood/Cabinet Screws	Main assembly fasteners for attaching base skids (P11), top panel (P1), bottom panel (P2), and outer frame
1 Pack	1 1/2″ #6 Star-Drive Wood/Cabinet Screws	Attaching support cleats (P10), stiffener bar (P9), and securing shelves (P7, P8) vertically down into cleats
1 Pack	1″ to 1-1/4″ Brad Nails or Wood Screws	Mounting the 3/4″ rear back panel (P3) around perimeter and center divider
1 Bit	1/8″ Countersink Drill Bit Combination	Pre-drilling pilot holes and recessing screw heads flush without splitting plywood

Lab-Grade Paint & Finishing

Qty	Item	Use	Priority
1 Quart	Latex Wood Primer	Seals raw plywood edges and surfaces prior to paint application	Mandatory
1 Gallon	Semi-Gloss or Gloss Interior Enamel Paint	Smooth, washable body finish resisting water and chemical splashes	Mandatory
1 Pack	120-Grit Sandpaper + 4″ Foam Rollers & Brushes	Surface sanding and smooth paint/sealer application	Mandatory

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2. Master Cut List

PartId	Part Name	Qty	Material	Cut Dimensions (W x D or H)	Function / Location
P1	Top Panel	1	3/4″ Plywood	92.00″ x 23.25″	Top exterior cover cap
P2	Bottom Panel	1	3/4″ Plywood	92.00″ x 23.25″	Bottom exterior base
P3	Back Panel	1	3/4″ Plywood	92.00″ x 24.00″	Full rear panel (locks box square; eliminates wall anchoring)
P4	Left Side Panel	1	3/4″ Plywood	22.50″ x 23.25″	Left outer wall
P5	Right Side Panel	1	3/4″ Plywood	22.50″ x 23.25″	Right outer wall
P6	Center Divider	1	3/4″ Plywood	22.50″ x 23.25″	Main vertical divider
P7	Long Shelf	1	3/4″ Plywood	60.25″ x 23.25″	Horizontal shelf for right section
P8	Small Shelves	3	3/4″ Plywood	29.50″ x 23.25″	Horizontal shelves for left section
P9	Shelf Stiffener	1	1x2 Pine Board	**56.00″ x 1.50″**	Centered under front edge of Long Shelf (P7), 2″ clearance each end for cleats
P10	Shelf Cleats	8	1x2 Pine Board	**20.00″ x 1.50″**	Support ledges mounted on inside side/divider walls
P11	Base Skids (Feet)	3	2x4 Lumber	**23.25″ x 3.50″**	Floor elevation skids mounted under bottom panel (P2)

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3. Sheet & Lumber Cutting Plan

```
PLYWOOD SHEET 1 (48" x 96" x 3/4")
+------------------------------------------------------------------------------------------+
|                         [P1] Top Panel: 92.0" x 23.25"                       | Remainder |
+------------------------------------------------------------------------------------------+
|                        [P2] Bottom Panel: 92.0" x 23.25"                     | Remainder |
+------------------------------------------------------------------------------------------+

PLYWOOD SHEET 2 (48" x 96" x 3/4")
+------------------------------------------------------------------------------------------+
|                         [P3] Back Panel: 92.0" x 24.00"                      | Remainder |
+------------------------------------------------------------------------------------------+
|       [P7] Long Shelf: 60.25" x 23.25"       |   [P8a] Small Shelf 1: 29.5"   | Remainder |
+------------------------------------------------------------------------------------------+

PLYWOOD SHEET 3 (48" x 96" x 3/4")
+------------------------------------------------------------------------------------------+
|   [P8b] Small Shelf 2: 29.5"  |   [P8c] Small Shelf 3: 29.5"  |         Scrap            |
+------------------------------------------------------------------------------------------+
| [P4] Left Side: 22.5"     | [P5] Right Side: 22.5"    | [P6] Center Divider: 22.5" | Scrap |
+------------------------------------------------------------------------------------------+

1x2 PINE BOARD #1 (Lip)   : Cut 1 piece @ 56.00" long [P9] (Stiffener Lip) + 40.00" Scrap
1x2 PINE BOARD #2 (Cleats): Cut 8 pieces @ 20.00" long [P10] (Support Cleats)
2x4 PINE BOARD (Skids)   : Cut 3 pieces @ 23.25" long [P11] (Floor Base Feet)
```

4. Visual Construction Diagrams

Diagram A: Long Shelf Stiffener Lip (End Cross-Section View)

Attaching a 1x2 pine board [P9] (cut to 56" length, centered with 2" clearance on each end) flush under the front edge of the long shelf [P7] creates an upside-down "L" beam that eliminates sagging under heavy chemistry gear. The centering prevents interference with cleat mounting areas.

```
                    SIDE VIEW (Looking down the side edge of the shelf)

                       Front Edge                              Back Edge
                            |                                       |
                            v                                       v
                    +----------------+----------------------------------------+
                    |           [P7] 3/4" Plywood Long Shelf                  |  <-- Horizontal top
                    +-------+--------+----------------------------------------+
                            |        |
                            | [P9]   |
                            | 1x2    |   <-- Vertical leg (hangs down 1.5 inches;
                            | Lip    |       aligned flush with front edge)
                            +--------+
```

Diagram B: Side Panel Support Cleat Placement (Side View of Interior Panels)

```
LEFT SECTION WALLS (Inside [P4] Left Side Panel & Left Face of [P6] Center Divider)
+--------------------------------------------------------------------------+ (Top Edge: 22.5" High)
|                                                                          |
|       [================== [P10] Cleat 3 ==================]              | <-- Top edge at 16.5"
|                                                                          |
|       [================== [P10] Cleat 2 ==================]              | <-- Top edge at 11.0"
|                                                                          |
|       [================== [P10] Cleat 1 ==================]              | <-- Top edge at 5.5"
|                                                                          |
+--------------------------------------------------------------------------+ (Bottom Edge)
| 3.25" Gap |<-------------------- 20.0" Long Cleat ---------------------->|
 (Front Edge)                                                             (Back Edge)

RIGHT SECTION WALLS (Right Face of [P6] Center Divider & Inside [P5] Right Side Panel)
+--------------------------------------------------------------------------+ (Top Edge: 22.5" High)
|                                                                          |
|       [================== [P10] Long Shelf Cleat =========]              | <-- Top edge at 14.5"
|                                                                          |
+--------------------------------------------------------------------------+ (Bottom Edge)
| 3.25" Gap |<-------------------- 20.0" Long Cleat ---------------------->|
```

Diagram C: Cabinet Assembly Layout - Front View with ALL Parts

Shows complete cabinet front view with BOTH left side panel (P4) and right side panel (P5).

```
FRONT VIEW: Looking at the cabinet from the front (facing the lab)

              ┌────────── [P1] TOP PANEL 92" x 23.25" ──────────┐
              │                                                 │
    ┌─────────┬──────────────┬──────────┬──────────┬────────────┬──────────┐
    │         │              │          │          │            │          │
    │  [P4]   │    [P8a]     │  [P6]    │  [P7]    │   [P9]     │  [P5]    │
    │ LEFT    │   Shelf 1    │ CENTER   │  LONG    │ STIFFENER  │ RIGHT    │
    │  SIDE   │ 29.5"x23.25" │ DIVIDER  │  SHELF   │ LIP UNDER  │ SIDE     │
    │  PANEL  │              │ 22.5"H   │ 60.25"x  │ (56" long) │ PANEL    │
    │ 22.5"H  │    [P10]     │          │ 23.25"   │            │ 22.5"H   │
    │         │    CLEATS    │ [P10]    │          │ [P10]      │          │
    │  [P10]  │ @ 5.5" H     │ CLEATS   │ [P10]    │ CLEATS     │          │
    │ CLEAT 1 │              │ @ 5.5" H │ @ 14.5"H │ @ 14.5" H  │          │
    │ 5.5" H  │    [P8b]     │          │          │            │          │
    │         │   Shelf 2    │ [P10]    │          │            │          │
    │  [P10]  │ 29.5"x23.25" │ CLEATS   │          │            │          │
    │ CLEAT 2 │              │ @ 11.0"H │          │            │          │
    │ 11.0" H │    [P10]     │          │          │            │          │
    │         │    CLEATS    │ [P10]    │          │            │          │
    │  [P10]  │ @ 16.5" H    │ CLEATS   │          │            │          │
    │ CLEAT 3 │              │ @ 16.5"H │          │            │          │
    │ 16.5" H │    [P8c]     │          │          │            │          │
    │         │   Shelf 3    │          │          │            │          │
    │         │ 29.5"x23.25" │          │          │            │          │
    │         │              │          │          │            │          │
    └─────────┴──────────────┴──────────┴──────────┴────────────┴──────────┘
              │                                                 │
              └────────── [P2] BOTTOM PANEL 92" x 23.25" ───────┘
                │                  │                  │
              [P11]              [P11]              [P11]  <- 2x4 Base Skids
             LEFT              CENTER             RIGHT

                    [P3] BACK PANEL 92" x 24" (mounts to rear)

KEY LAYOUT:
═══════════
LEFT:   P4 (Left Side Panel) - 22.5"x23.25" on far left
        P8a, P8b, P8c (3 Small Shelves) - 29.5" wide
        P10 Cleats (3 total) at 5.5", 11.0", 16.5" heights

CENTER: P6 (Center Divider) - 22.5"x23.25" in middle
        P10 Cleats (3 total) on BOTH faces at same heights

RIGHT:  P5 (Right Side Panel) - 22.5"x23.25" on far right
        P7 (Long Shelf) - 60.25"x23.25"
        P9 (Stiffener Lip) - 56" long, under P7 front edge
        P10 Cleat (1 total) at 14.5" height

BOTH SIDES: P1 (Top), P2 (Bottom), P3 (Back), P11 (3 Skids)
```

5. Step-by-Step Build Instructions

Step 1: Prepare Long Shelf Stiffener Lip

1. Lay the **Long Shelf [P7]** (60.25″ x 23.25″) flat on your workbench bottom-side up.
2. Position the 1x2 pine board **[P9]** (cut to **56″ long**) centered under P7 with **approximately 2″ clearance on each end** to avoid interference with cleat mounting areas on vertical panels (P5, P6).
3. Align **P9** on its narrow edge **flush against the front edge** on the underside of **P7** (forming an inverted "L" beam profile) with proper centering for cleat clearance.
4. Apply wood glue along the contact joint, clamp in place, and drive **1-1/4″ to 1-1/2″ finish nails** (or pre-drill with a 1/8″ countersink bit and drive 1-1/4″ wood screws) vertically down through the top face of **P7** into **P9** every 6 to 8 inches. Wipe away glue squeeze-out.

VISUAL: Step 1 - How It Looks (Side View)

```
BEFORE: P7 and P9 separated
   ┌──────────────────────────────────┐
   │  [P7] Long Shelf (bottom-side up)│  60.25" x 23.25"
   └──────────────────────────────────┘

     ┌────────────────────┐
     │ [P9] 1x2 Stiffener │  56" long (centered)
     └────────────────────┘

AFTER ASSEMBLY: P9 glued and clamped to P7
   ┌──────────────────────────────────────────┐  TOP (shows screw holes)
   │  [P7] Long Shelf (3/4" plywood)          │  60.25" x 23.25"
   │  ●      ●      ●      ●      ●          │ Fasteners: ● ● ● (6-8" spacing)
   │  ●      ●      ●      ●      ●          │
   ├──────────────────────────────────────────┤
   │                  │                       │
   │ ↓CLAMP LOCATION  │     ↓CLAMP            │
   │  └───────────────┘                       │  UNDER: Glued & clamped P9
   │                                          │         (centered with 2" gap each end)
   │  ┌────────────────────────────────────┐  │
   │  │ [P9] 1x2 Stiffener Lip (1.5" tall) │  │
   │  └────────────────────────────────────┘  │
   │  ↑CLAMP LOCATION                         │
   │  └──────────────────┘                    │

CLAMP PLACEMENT DETAIL:
  Use 2-3 bar clamps or C-clamps (18"-24" capacity)
  Position clamps evenly along 56" length

  Front clamp         Mid clamp            Back clamp
  ↓                   ↓                     ↓
  ╔═════════════════════════════════════════════════════╗
  ║ [P7 + P9 Assembly] - Clamped to workbench           ║
  ║ Hold firmly while glue cures (15-30 minutes)        ║
  ╚═════════════════════════════════════════════════════╝

RESULT: Inverted "L" beam prevents sagging
   ┌────────────┐
   │ [P7] Shelf │  <- Stiffens front edge
   │ ┌────────┐ │
   │ │ [P9]   │ │  <- Extends down 1.5"
   │ └────────┘ │
   └────────────┘
```

Step 2: Attach 2x4 Base Skids (Floor Water & Moisture Protection)

1. Lay the **Bottom Panel [P2]** (92″ x 23.25″) flat on the floor face down.
2. Cut three 23.25″ long 2x4 pieces **[P11]** (Base Skids). Place one flush at the far left edge, one in the exact middle under the center divider line, and one flush at the far right edge (running front-to-back).
3. Apply wood glue between **P11** skids and **P2** plywood. Pre-drill with your 1/8″ countersink bit and drive **2″ #8 wood screws** through **P2** into **P11]** (4 screws per skid).
4. *Function:* Elevates **P2** 1.5″ off damp chemistry lab floors, preventing mopping water, puddles, and chemical cleaners from soaking into the wood core while making the unit easier to slide into place.

VISUAL: Step 2 - How It Looks (Top-Down View)

```
BEFORE: P2 panel with 3 skids ready to install

P2 Panel (92" x 23.25")
┌──────────────────────────────────────────────────────────┐
│                                                          │
│  [P11]        [P11]             [P11]                    │
│  LEFT SKID    CENTER SKID       RIGHT SKID               │
│  (not yet      (not yet          (not yet                │
│   attached)    attached)          attached)              │
│                                                          │
└──────────────────────────────────────────────────────────┘

SIDE VIEW: Positioning skids on P2

  FLOOR
  ═════════════════════════════════════════════════════════

  ┌────────────────────────────────────────────────────────┐
  │ [P2] Bottom Panel - laid flat, top-side showing        │
  │                                                        │
  │ Glue application here (continuous bead along top)      │
  │ ↓↓↓↓↓↓↓↓↓↓↓↓↓↓↓↓↓↓↓↓↓↓↓↓↓↓↓↓↓↓↓↓↓↓↓↓↓↓↓↓↓↓↓↓↓↓↓↓ │
  │                                                        │
  │  ╔═════════╗                                           │
  │  ║ [P11]   ║     ╔═════════╗         ╔═════════╗       │
  │  ║ Left    ║     ║[P11]    ║         ║[P11]    ║       │
  │  ║ Skid    ║     ║Center   ║         ║Right    ║       │
  │  ╚═════════╝     ║ Skid    ║         ║ Skid    ║       │
  │  ↑               ╚═════════╝         ╚═════════╝       │
  │  (at edge)       (at 46")            (at edge)         │
  └────────────────────────────────────────────────────────┘

  After fastening:

  ╔════════════════════════════════════════════════════════╗
  ║ SCREW PLACEMENT (4 screws per skid, top of P2 face)    ║
  ║                                                        ║
  ║ Each skid: [●] 2 screws at front end                   ║
  ║            [●] 2 screws at back end                    ║
  ║                                                        ║
  ║ Use: 2" #8 Star-Drive Screws (pre-drill first!)        ║
  ║ Countersink so heads sit flush                         ║
  ╚════════════════════════════════════════════════════════╝

CLAMP USAGE (optional but helpful):
  Use 2-3 bar clamps to hold P11 skids firmly against P2
  while driving screws. This prevents shifting.

RESULT: Elevated base for moisture protection

      Floor level
      ═════════════════════════════════════════
      ▲ ▲ ▲ ▲ ▲ ▲ ▲ ▲ ▲ ▲ ▲ ▲ ▲ ▲ ▲ ▲ ▲ ▲ ▲
    ┌────────────────────────────────────────────────┐
    │ [P2] Bottom Panel                              │
    │ ┌──────┐          ┌─────────┐          ┌──────┐│
    │ │[P11] │          │  [P11]  │          │[P11] ││
    │ │      │          │         │          │      ││
    │ └──────┘          └─────────┘          └──────┘│
    │                                                │
    └────────────────────────────────────────────────┘

     1.5" gap <- Water/moisture protected!
```

Step 3: Mount Side Support Cleats (The Easy Assembly Method)

1. Measure from the bottom edge of your vertical wall panels (**P4**, **P5**, **P6** inside faces only-**no wall mounting to room walls**) and draw horizontal guidelines:

- **Left Side Panel [P4] & Left Face of Center Divider [P6]:** Draw lines at **5.5"**, **11.0"**, and **16.5"** from the bottom.
- **Right Side Panel [P5] & Right Face of Center Divider [P6]:** Draw a line at **14.5"** from the bottom.

2. Position a 20" long 1x2 cleat **[P10]** *just below each line*, pushing the cleat flush against the rear edge (leaving a 3.25" space at the front).
3. Apply wood glue behind each cleat **[P10]**, pre-drill pilot holes, and secure with **1-1/4" finish nails or 1-1/4" wood screws** (3 fasteners per cleat).

VISUAL: Step 3 - How It Looks (Vertical Panel from Inside)

```
BEFORE: Panel with height marks, cleats ready to mount

  Vertical Panel (P4, P5, or P6) - 22.5"H x 23.25"D

  22.5" ─────────────────────────────────────────
         (Top edge)

  16.5" ─ <- Draw line (mark with pencil)

  11.0" ─ <- Draw line (mark with pencil)

   5.5" ─ <- Draw line (mark with pencil)

   0.0" ─────────────────────────────────────────
         (Bottom edge)

AFTER: Cleats mounted at each height

         FRONT                    BACK

  22.5"  ┌───────────────────────────────────┐
  16.5"  │                                   │
         │  ╔══════════════════════════════╗ │  <- Top of cleat @ 16.5"
         │  ║ [P10] CLEAT #3               ║ │  <- Flush to back edge
         │  ║ 1x2 Pine, 20" long           ║ │     Gap at front ↑
         │  ╚══════════════════════════════╝ │     (3.25" empty)
         │  ●                            ●   │
         │ (fastener)              (fastener)│
  11.0"  │                                   │
         │  ╔══════════════════════════════╗ │  <- Top of cleat @ 11.0"
         │  ║ [P10] CLEAT #2               ║ │
         │  ╚══════════════════════════════╝ │
         │  ●      ●      ●                │ │
   5.5"  │                                   │
         │  ╔══════════════════════════════╗ │  <- Top of cleat @ 5.5"
         │  ║ [P10] CLEAT #1               ║ │
         │  ╚══════════════════════════════╝ │
         │  ●                            ●   │
   0.0"  └───────────────────────────────────┘

  <- 3.25" gap    20" cleat    Flush back
     (front)      length        edge

FASTENER PLACEMENT (3 per cleat):

  ╔═════════════════════════════════════════╗
  ║ CLEAT (20" long 1x2)                    ║
  ●                ●                    ●
  ↑                ↑                    ↑
  Left end      Center (10")         Right end
  (2" in)        (middle)             (2" in)

  Use: 1.25" finish nails OR 1.25" wood screws
       Drive HORIZONTALLY through cleat into panel

CLAMP USAGE:

  Use spring clamps or quick-grip clamps to hold
  cleat flush against panel while fastening:

  ┌─────────────────────────────────────────┐
  │ Panel (back surface)                    │
  │ ○ <- clamp here (gently)                │
  │                                         │
  │ ╔═════════════════════════════════════╗ │
  │ ║ [P10] CLEAT                         ║ │
  │ ║ (1.5" tall x 1.5" deep x 20" long)  ║ │
  │ ╚═════════════════════════════════════╝ │
  │                                         │
  │ ○ <- clamp here (gently)                │
  └─────────────────────────────────────────┘

REPEAT for all 8 cleats:
  - Left section (P4): 3 cleats
  - Center (P6): 3 cleats (both faces!)
  - Right section (P5): 2 cleats
```

Step 4: Assemble Interior Shelves & Frame (Dual-Fastening Method)

1. Stand the **Left Side Panel [P4]**, **Center Divider [P6]**, and **Right Side Panel [P5]** vertically.
2. Slide the 3 **Small Shelves [P8]** (29.5") and 1 **Long Shelf [P7]** (60.25") into position so they rest directly on top of the mounted interior support cleats **[P10]**.
3. Apply wood glue along all cleat and side wall contact joints.
4. **Fasten Shelves (Dual-Fastening Rule):**

- **Vertical Cleat Fastening:** Drive **1-1/4″ to 1-1/2″ finish nails** (or 1-1/4″ screws) vertically down through the top face of shelves (**P7**, **P8**) into the 1x2 support cleats **[P10]** below (3 fasteners per cleat).
- **Horizontal Side Fastening:** Drive **1-1/2″ to 2″ finish nails** (or 1-5/8″ trim screws) horizontally through exterior side panels (**P4**, **P5**) and divider **[P6]** directly into the center edge cores of the shelf boards (2-3 fasteners per shelf side).

VISUAL: Step 4 - How It Looks (Shelves Resting on Cleats)

```
BEFORE: Vertical panels standing, shelves ready to slide in

  P4 (L)      P6 (C)       P5 (R)
    ║            ║              ║
    ║            ║              ║
    ║  [cleat]   ║  [cleat]     ║
    ║   ════════ ║ ════════     ║
    ║            ║              ║
    ║  [cleat]   ║  [cleat]     ║
    ║   ════════ ║ ════════     ║
    ║            ║              ║
    ║  [cleat]   ║  [cleat]     ║
    ║   ════════ ║ ════════     ║

AFTER: Shelves slid onto cleats

SIDE VIEW: Looking at one shelf on its cleats

  BACK (wall)                     FRONT (lab)
  ║                               ║
  ║ [Panel back surface]          ║
  ║ ╔═════════════════════════════╝
  ║ ║ [P10] CLEAT
  ║ ║══════════════════════════════════════╗
  ║ ║                                      ║
  ║ └──────────────────────────────────────┘
  ║                                        │
  ║ ╔════════════════════════════════════╗ │
  ║ ║ [SHELF] - resting on cleat         ║ │ <- Shelf sits on top
  ║ ║ (Glue applied to top of cleat)     ║ │
  ║ ╚════════════════════════════════════╝ │
  ║           ↑                            ║
  ║      Cleat top surface                 ║
  ║      supports shelf                    ║

TOP-DOWN VIEW: Looking down at one shelf with fastener locations

  ┌──────────────────────────────────────┐
  │ [SHELF TOP SURFACE] - showing         │
  │ fastener holes                        │
  │                                       │
  │ ● ● ● [shelf spans across cleats] │ <- Vertical fasteners
  │                                       │    (down through shelf)
  │                                       │
  │ ↓      ↓      ↓                       │
  │ L      C      R                       │
  │ shelf  shelf  shelf                   │
  │ end    center end                     │
  │                                       │
  │ (3 fasteners per cleat, spaced        │
  │  left end, center 10", right end)     │
  └──────────────────────────────────────┘

FASTENING METHOD #1: VERTICAL (through shelf into cleat)

  ┌──────────────────────────────────┐  SHELF TOP VIEW
  │          (●) Fastener hole        │
  │                                   │
  │ [==== SHELF ====]                 │  Fasteners go down through shelf
  │ ●             ●              ●    │  ↓ into cleats below
  │ L      C      R  (3 per cleat)    │
  └──────────────────────────────────┘

  Use: 1.25"-1.5" finish nails OR 1.25" wood screws
  Pre-drill with 1/8" countersink bit first

FASTENING METHOD #2: HORIZONTAL (through side panels into shelf)

  SIDE VIEW showing horizontal fasteners

  ┌─────────────────────────────────┐
  │ [Side Panel] P4, P5, or P6       │
  │                                  │
  │ ●                                │ <- Fastener through panel
  │ ●                                │      into shelf edge
  │ ●                                │
  │ ↓ 2-3 fasteners per shelf side   │
  │                                  │
  │ ═════════════════════════════    │
  │    (Shelf end/edge core)         │
  └─────────────────────────────────┘

  Use: 1.5"-2" finish nails OR 1.625" trim screws
  Drive HORIZONTALLY through panel into shelf edge center

CLAMP USAGE (optional but helpful):

  Use spring clamps to hold shelves level while fastening:

  ┌─────────────────────────────────┐
```

```
  │ ○ clamp here                │  <- Light pressure
  │ ┌──────────────────────────┐│
  │ │ [Shelf]                  ││
  │ └──────────────────────────┘│
  │ ○ clamp here                │  <- Light pressure
  └─────────────────────────────┘

RESULT: Shelves secure on cleats, ready for final assembly

  All shelves properly seated:
  [OK] P8a at 5.5" (small shelf 1)
  [OK] P8b at 11.0" (small shelf 2)
  [OK] P8c at 16.5" (small shelf 3)
  [OK] P7 at 14.5" (long shelf - right section)
```

Step 5: Attach Top, Bottom & Back Panels (Free-Standing Rigidity)

1. Apply wood glue to all top edges of the assembled vertical walls (**P4**, **P5**, **P6**).
2. Place the **Top Panel [P1]** (92″ x 23.25″) squarely on top, pre-drill, and secure with **2″ wood screws** into vertical panel edges (3 screws per wall).
3. Apply glue to all bottom edges, set the frame onto the **Bottom Panel [P2]** (with attached **P11** skids), and secure with **2″ wood screws** driven from inside down into **P2**.
4. Lay the entire cabinet face down. Measure diagonally corner-to-corner to verify the frame is perfectly square.
5. Apply wood glue around the full rear perimeter and center divider edge.
6. Set the **Back Panel [P3]** (92″ x 24″) flush over the rear face. Pre-drill and drive **1″ to 1-1/4″ brad nails or screws** every 6 to 8 inches around the perimeter and center divider. *This rear panel locks the frame square and provides the structural stability required so the shelf stays completely rigid without any wall mounting.*

VISUAL: Step 5a - How It Looks (Attaching Top Panel [P1])

```
BEFORE: Vertical frame standing, P1 ready to attach

  P1 (Top Panel)
  ┌────────────────────────────────────┐
  │ 92" x 23.25"                       │
  └────────────────────────────────────┘
        ↓ (place on top)

  ┌──────────┬─────────┬──────────┐
  │ P4       │ P6      │ P5       │  <- Vertical panels
  │ (cleat)  │ (cleat) │ (cleat)  │
  │ (shelf)  │ (shelf) │ (shelf)  │
  └──────────┴─────────┴──────────┘

AFTER: P1 glued and clamped to top of frame

  ┌────────────────────────────────────┐
  │ [P1] TOP PANEL - clamped in place  │
  │ ●              ●              ●    │ <- Pre-drilled screw locations
  │ Fasteners: 3 screws per wall edge  │    (P4, P6, P5)
  └────────────────────────────────────┘
        │          │          │
      clamp      clamp      clamp
      (2-3)      (1-2)      (2-3)
        │          │          │

CLAMP ARRANGEMENT (looking from above):

  Front edge clamp
           ↓
  ╔════════●═════════╗
  ║    [P1 TOP]     ║
  ║   ┌────┬────┐    ║
  ║   │ P4 │ P6 │ P5 ║
  ║   └────┴────┘    ║
  ╚════════●═════════╝
           ↑
      Back edge clamp

  Hold P1 firmly while driving screws
  2" #8 Star-Drive Wood Screws (countersink)

RESULT: Top panel locks frame together

  ┌──────────────────────────────────┐
  │ [P1] - Firmly attached to top    │
  │ of vertical frame                │
  └──────────────────────────────────┘
  │          │          │
  └──────────┴──────────┘
     Solid top!
```

VISUAL: Step 5b - How It Looks (Attaching Bottom Panel [P2] & Squaring Cabinet)

```
BEFORE: Frame with P1 on top, ready for P2

        ┌─────────────────────────┐
        │ [P1] (Top – attached)   │
        │                         │
        │ [Vertical frame]        │
        │                         │
        └─────────────────────────┘

        ┌─────────────────────────┐
        │ [P2] – Ready to attach  │
        │ (with P11 skids below)  │
        └─────────────────────────┘


AFTER: P2 attached, cabinet face-down on workbench

  Workbench
  ═════════════════════════════════════════════════

  ┌─────────────────────────────────────────────────┐
  │ [P3 – Back Panel – NOT YET attached]            │
  │                                                 │
  │ ╔═════════════════════════════════╗             │
  │ ║ CABINET FACE-DOWN (upside down) ║             │
  │ ║ with P2 (bottom) now facing up  ║             │
  │ ║                                 ║             │
  │ ║ P1 is now on bottom (against    ║             │
  │ ║ the workbench)                  ║             │
  │ ║                                 ║             │
  │ ║ [P2] Bottom panel attached      ║             │
  │ ║ with P11 skids visible below    ║             │
  │ ╚═════════════════════════════════╝             │
  │                                                 │
  └─────────────────────────────────────────────────┘


CRITICAL: SQUARING THE CABINET (measure diagonals)

  Diagonal #1              Diagonal #2
  TL -> BR                  TR -> BL
   ↘                       ↙
  ┌───────────────────────┐
  │ <-  Measure this      │ │ <- Measure this ->
  │      diagonal ->      │ │
  ├───────────────────────┤
  │      <- Measure       │ │ <- Measure ->
  │      this diagonal    │
  └───────────────────────┘
   ↙                       ↘
  BL -> TR                  BR -> TL

  IF DIAGONALS ARE EQUAL: [OK] Cabinet is SQUARE
  IF NOT EQUAL: Loosen fasteners and adjust frame

  Both diagonals should be within 1/4" of each other

  Target: Cabinet is perfectly rectangular before
          attaching back panel!


ATTACHING P2 (Bottom Panel):

  ╔════════════════════════════════════╗
  ║ [P2] Bottom Panel – Top side showing ║
  ║ (with P11 skids underneath)        ║
  ║                                    ║
  ║ ● ● ● Fastener locations           ║
  ║ ┌─────────────────────────────────┐ ║
  ║ │ [Vertical panels above]         │ ║
  ║ │ Screws driven from inside:      │ ║
  ║ │ 3 screws per panel (P4, P6, P5) │ ║
  ║ └─────────────────────────────────┘ ║
  ║                                    ║
  ║ 2" #8 Star-Drive Wood Screws       ║
  ║ (pre-drill pilot holes first!)     ║
  ╚════════════════════════════════════╝

RESULT: Cabinet now complete except for P3 back panel

  All 4 sides assembled:
  [OK] P1 (top) – attached
  [OK] P2 (bottom) – attached
  [OK] P4, P6, P5 (sides/divider) – in place
  [OK] P3 (back) – READY TO ATTACH

  Cabinet is rigid but NOT LOCKED SQUARE until P3
  is attached!
```

VISUAL: Step 5c - How It Looks (Attaching Back Panel [P3])

```
BEFORE: Cabinet ready for back panel

  Cabinet positioned face-down on workbench
  (ready for P3 to be attached)

  ╔════════════════════════════════════════╗
  ║ [P3 - Back Panel 92" x 24"]            ║
  ║ (ready to attach)                      ║
  ║ ↓                                      ║ ║
  ║ ┌──────────────────────────────────┐   ║
  ║ │ [Empty rear surface]             │   ║
  ║ │ (interior of cabinet)            │   ║
  ║ │                                  │   ║
  ║ │ Ready for P3 to mount flush      │   ║
  ║ └──────────────────────────────────┘   ║
  ║                                        ║
  ╚════════════════════════════════════════╝

AFTER: P3 attached (rear locked)

  ╔════════════════════════════════════════╗
  ║ [P3] Back Panel - ATTACHED              ║
  ║ Showing fastener pattern around edge:   ║
  ║                                          ║
  ║ ● ● ● ● ● ● ● (top edge - every 6-8") ║
  ║ ┌──────────────────────────────────┐   ║
  ║ │ [Cabinet interior]               │   ║
  ║ │ ●      ●      ●                  │   ║ L
  ║ │ (center divider P6 fasteners)    │   │ e
  ║ │ ●      ●      ●                  │   │ f
  ║ └──────────────────────────────────┘   │ t
  ║ ● ● ● ● ● ● ●  (bottom - every 6-8")  │
  ║                                        │ E
  ║       (Right edge fasteners shown)     │ d
  ║       ● ● ● ● ● ● ●                   │ g
  ║       (Left edge fasteners shown)      │ e
  ║       ● ● ● ● ● ● ●                   │
  ╚════════════════════════════════════════╝

FASTENER PATTERN:

  Every 6-8 inches around entire perimeter:
  [OK] Top edge (92" wide) = ~12 fasteners
  [OK] Bottom edge (92" wide) = ~12 fasteners
  [OK] Left edge (24" high) = ~4 fasteners
  [OK] Right edge (24" high) = ~4 fasteners
  [OK] Center divider (P6) back side = ~4 fasteners

  TOTAL: 20-24 fasteners holding P3

  Use: 1" to 1.25" brad nails OR 1" wood screws
       (pre-drill pilot holes)

CRITICAL: Back panel LOCKS cabinet rigid!

  Once P3 is fully fastened:
  [OK] Cabinet cannot rack (shift out of square)
  [OK] Cabinet is rigid without wall mounting
  [OK] Back panel prevents any twisting
  [OK] Frame is permanently locked square
```

> **NOTE** When fully assembled, three sheets of 3/4" plywood weigh approximately 160 lbs! Assemble this shelf unit inside or right next to the chemistry lab room where it will stay. Always have 3 to 4 Scouts work together when lifting, moving, or flipping the cabinet frame.